

z-test

$$\text{test score} > 70 \approx \mu_0$$

$$\sigma = 4$$

$$\text{student score} = [72, 65, 70, 68, 75, 80, 78, 74, 69, 71]$$

$$\bar{x} = 722/10 = 72.2$$

✓ H_0 : Sudhansu's claim is correct > 70

H_1 : Sudhansu's claim is Not correct < 70

$$\alpha = 0.05$$

z-test

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

$$z = \underline{\underline{1.74}}$$

$$\bar{x} = 72.2$$

$$\mu_0 = 70$$

$$\sigma = 4$$

$$n = 10$$

$$z = \underline{1.74} = \left(\underline{1.7} + 0.04 \right) = \underline{1.74}$$

$$\text{critical } z \text{ value} = 1.645$$

$$z \geq z_c \rightarrow \underline{\text{Reject } H_0}$$

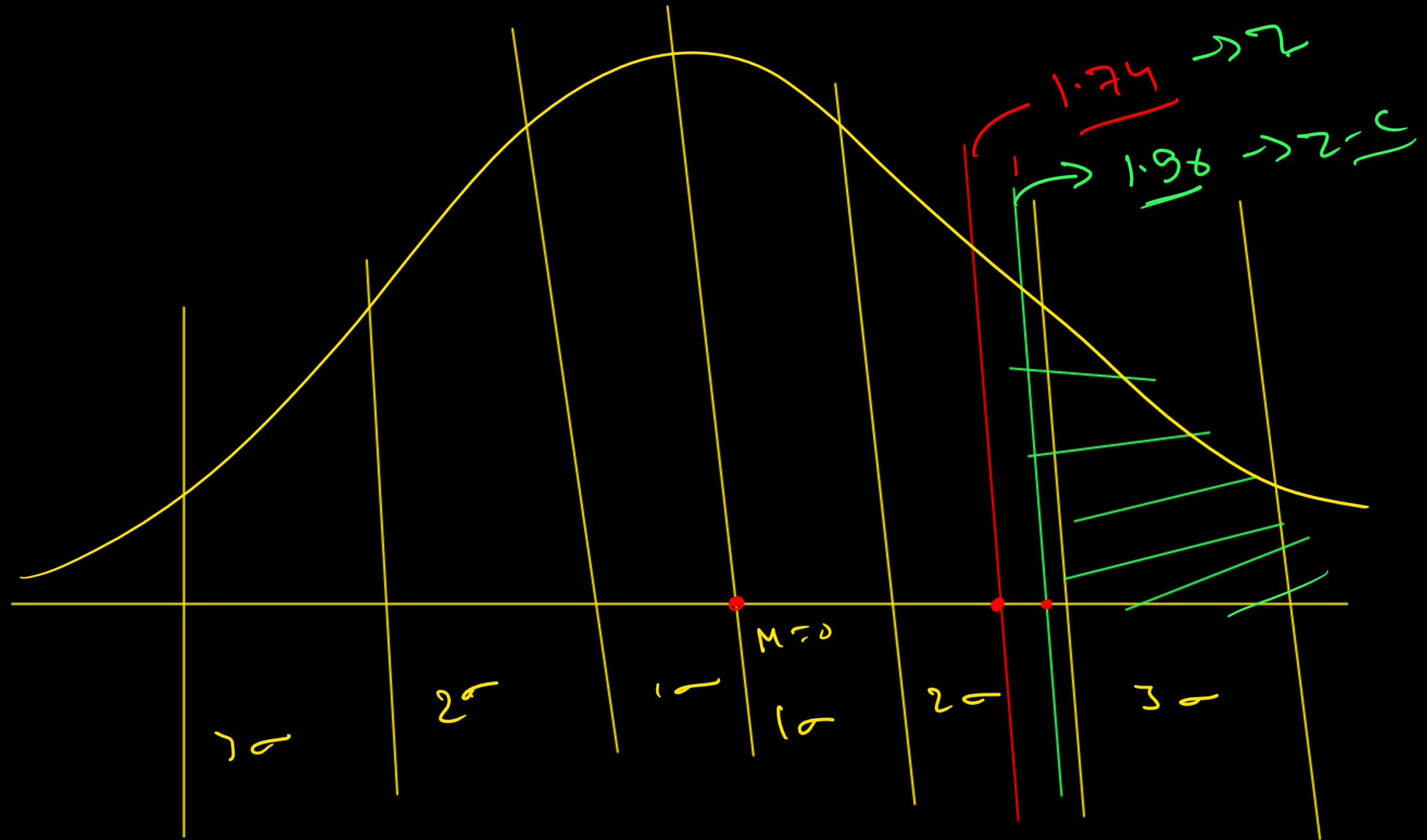
$$z < z_c \rightarrow \text{Accept } H_0$$

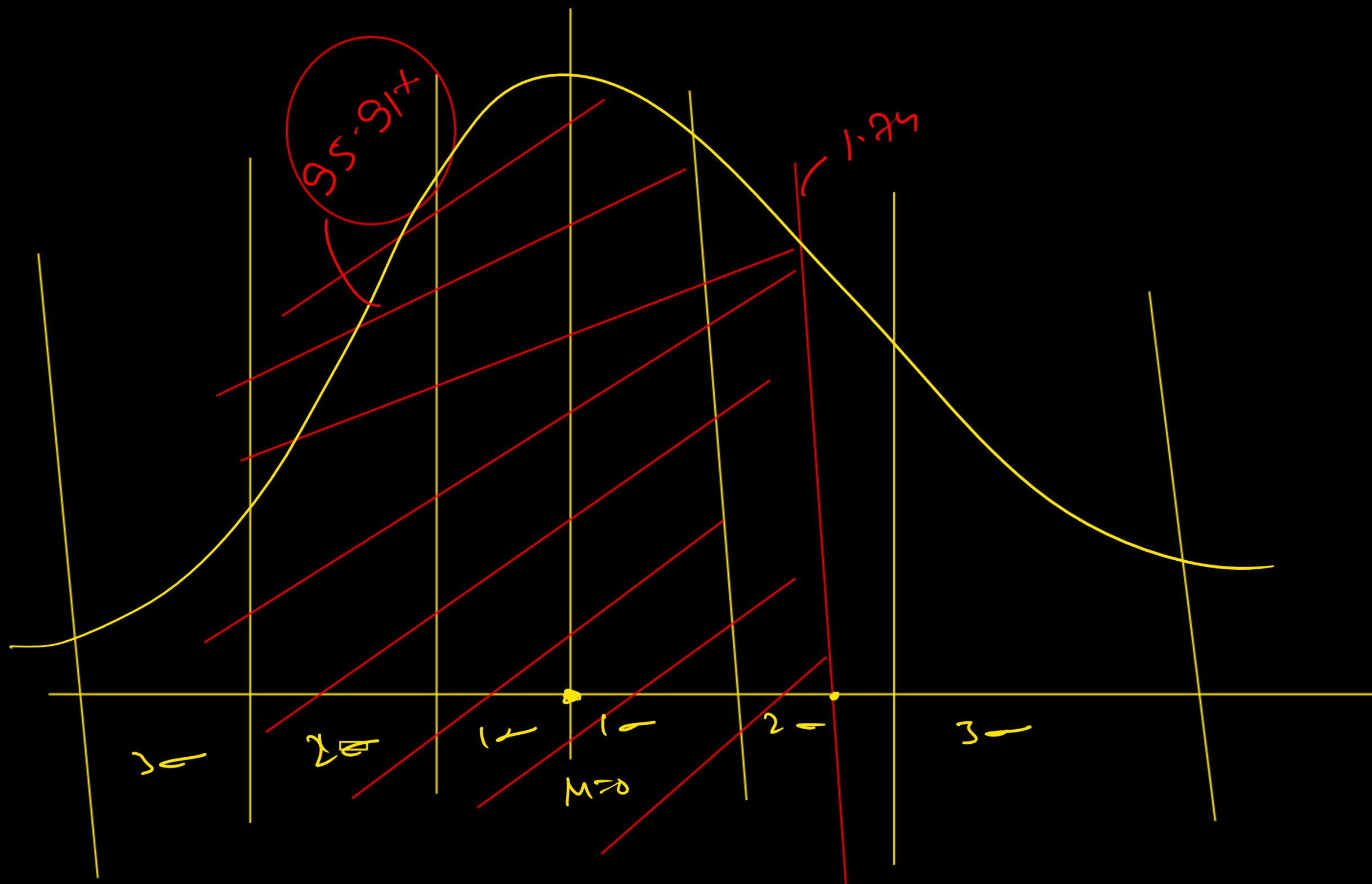
$$1.74 \geq 1.645$$

$$\underline{\underline{z_c = 1.96}}$$

$$(1.74 < 1.96)$$

$$z = 1.74$$





Chi-Square test



	Like math	Don't like math	Row total
→ Boys	12 ⁰	8 ¹	20
→ Girls	8 ¹	12 ⁰	20
→ Col total	20	20	40

H_0

There is no relation between Gender & liking math.

H_1

There is a relation between Gender & liking math.

$$\text{Chi-Square } (\chi^2) = \sum \frac{(\text{observed} - \text{Expected})^2}{\text{Expected}}$$

$$\text{Expected Count} = \frac{\text{Row total} \times \text{Col total}}{\text{Grand total}}$$

$$\begin{aligned}
 &= \\
 \checkmark - \text{Row total (Boys)} &= 20 \checkmark \\
 \checkmark - \text{Row total (Girl)} &= 20 \\
 \checkmark - \text{Col total (like math)} &= 20 \checkmark \\
 \checkmark - \text{Col total (Not like math)} &= 20 \\
 \text{Grand total} &= 40
 \end{aligned}$$

$$E_{B,L} = \frac{20 \times 20}{40} = 10$$

$$E_{B,DL} = \frac{20 \times 20}{40} = 10$$

$$E_{C,L} = \frac{20 \times 20}{40} = 10$$

$$E_{C,DL} = \frac{20 \times 20}{40} = 10$$

Bay like math

$$D = 12$$

$$E = 10$$

B, DL math

$$D = 8$$

$$E = 10$$

$$(D - E)^2 = (8 - 10)^2 = (-2)^2 = 4 = 0.4$$

$$\chi^2_{B1} = \frac{(0 - E)^2}{E} = \frac{2^2}{10} = 0.4$$

$$\frac{1}{E}$$

$$C, L \text{ mod}$$

$$O = 8$$

$$E = 10$$

$$\frac{(O - E)^2}{E} = \frac{8 - 10}{10} = \frac{4}{10} = 0.4$$

$$C, D, L \text{ mod}$$

$$O = 12$$

$$E = 10$$

$$\frac{(O - E)^2}{E} = \frac{2^2}{10} = 0.4$$

$$\chi^2 = 0.4 + 0.4 + 0.4 + 0.4 = 1.6$$

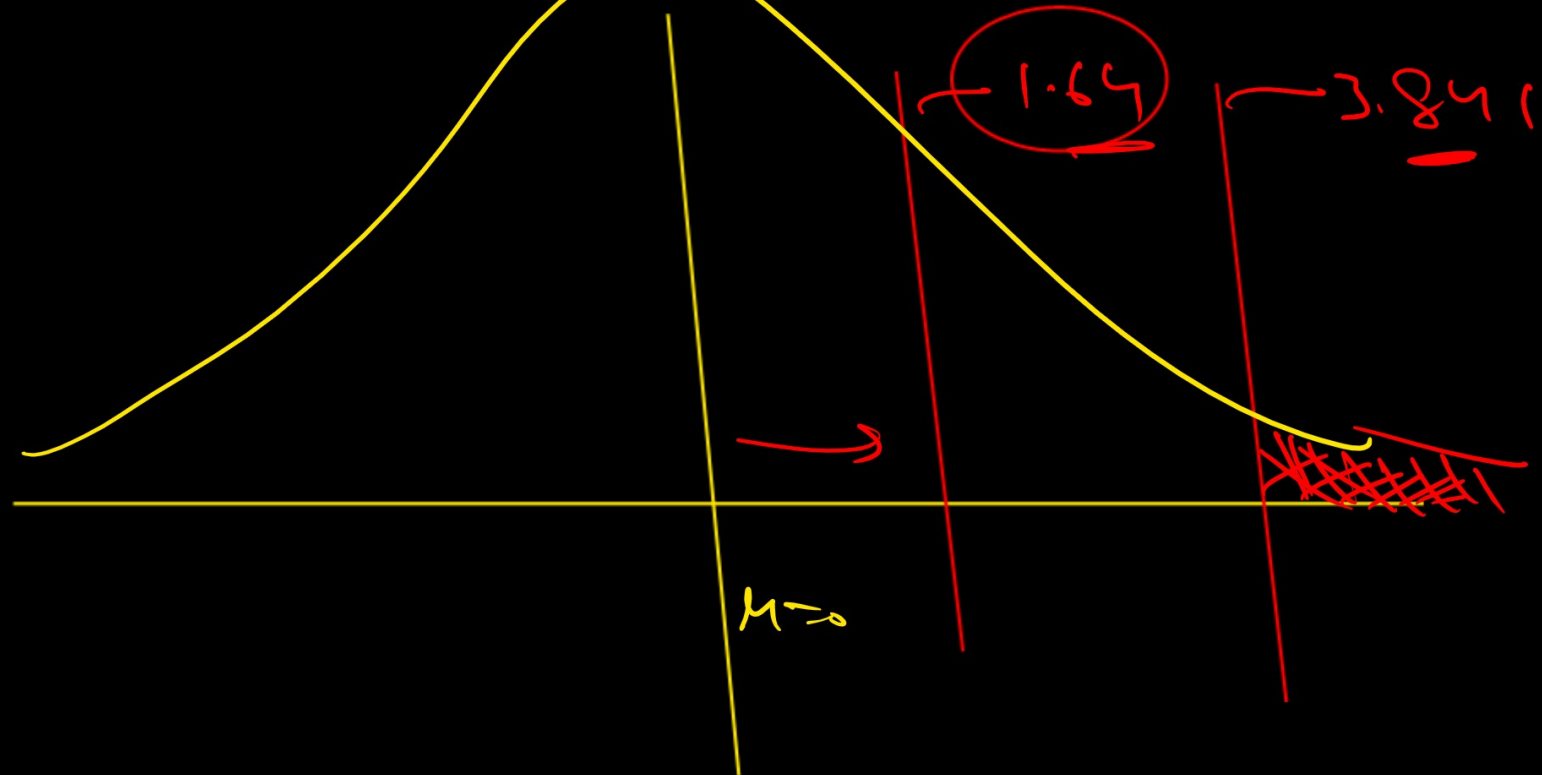
$$df = (n - 1) \times (c - 1)$$

$$= (2 - 1) \times (2 - 1)$$

= 1

$$\chi^2 = 1.6$$

$$\text{Critical } \chi^2 = 3.841$$



ANOVA

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✓ Method A → Traditional style. ✓

✓ Method B → Practical manner ✓

✓ Method C → Lecture + Practical ✓

Score Student	8 ✓ Method A	4.4 ✓ Method B	✓ 8.6 Method C	
1	8	4	9	
2	9	5	7	
3	6	3	8	
4	7	4	10	
5	10	6	9	

Mean Calculation:

$$\overline{\text{Method A}} = 8$$

$$\overline{\text{Method B}} = 4.4$$

$$\overline{\text{Method C}} = 8.6$$

$$\overline{\text{Grand Avg}} = 7.0$$

$$\bar{X} = \frac{40 + 22 + 43}{15} = \frac{105}{15} = 7.0$$

$$\bar{X} = 7.0$$

Total Number of Student = 15

H₀: All teaching method have same impact

H₁: At least one method is diff from rest

$$\bar{X}_A = 8.0$$

$$\bar{X}_B = 4.4$$

$$\bar{X}_C = 8.6$$

$$\bar{X} = 7$$

$$SS_{within} = \sum (x - \bar{x}_{group})^2$$

$$SS_{A, within} = (8-8)^2 + (9-8)^2 + (6-8)^2 + (7-8)^2 + (10-8)^2$$

$$= 0 + 1 + 4 + 1 + 4 = 10$$

$$SS_{B, within} = (4-4.4)^2 + (5-4.4)^2 + (3-4.4)^2 + (4-4.4)^2 + (6-4.4)^2$$

$$= 5.2$$

$$SS_{C, within} = 5.2$$

$$SS_{within} = SS_A + SS_B + SS_C$$

$$= 10 + 5.2 + 5.2 = 20.4$$

$$SS_{\text{between}} = \sum n_i (\bar{X}_i - \bar{X})^2$$

$$= 5 \times (8 - 7)^2 + 5 (4.4 - 7)^2 + 5 \times (8.6 - 7)^2$$

$$= 51.6$$

$$SS_{\text{total}} = \sum (x - \bar{X})^2$$

$$= 72.0$$

$$SS_{\text{within}} = 20.4$$

$$SS_{\text{between}} = 51.6$$

$$SS_{\text{total}} = 72$$

$$SS_{\text{total}} \approx SS_{\text{within}} + SS_{\text{between}}$$

$$20.4 + 51.6$$

$$72 \approx 72$$

F-table

df $\rightarrow$

$$\text{df between} = k - 1 = 3 - 1 = 2$$

No. of groups $\nearrow$

$$\text{df within} = N - k = 15 - 3 = 12$$

No. of observations $\nwarrow$

$$\text{df total} = N - 1 = 15 - 1 = 14$$

$$\text{MS between} = \frac{SS \text{ between}}{\text{df between}} = \frac{51.6}{2} = 25.8$$

$$F = \frac{\text{MS between}}{\text{MS within}}$$

MS within

$$\text{MS within} = \frac{SS \text{ within}}{\text{df within}} = \frac{20.4}{12} = 1.7$$

$$= \frac{25.8}{1.7} \approx 15.18$$

